

Day 10 — Final Assessment + Project Showcase

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the Nikhilam multiplication method via a summative quiz and a 3-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric per pair (`assessments/rubric.md`).
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- Brief: “Quiz first (15 min), presentations after.”

Part 1: Summative quiz (15 min)

- Distribute `quizzes/summative.md`. Closed notebook.
- Quiz covers:
 - Complement / excess fluency (timed).
 - Both-below-100 (2 problems).
 - Both-above-100 (1 problem).
 - Mixed-sign (1 problem).
 - Near-1000 (1 problem).
 - Decide-which-method (3 problems).
 - One short-answer “why does it work” (algebra).
- Collect at 15-min mark.

Part 2: Project presentations (25 min)

- 3 min presentation + 1 min Q&A per pair (4 min total).
- ~6 pairs fit in 25 min. **Reserve next class for remaining pairs** if needed.
- During each:
 - Teacher marks the rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): *“What’s one thing you’ll keep using from Nikhilam? And one thing you’ll keep using long multiplication for?”*
- Reflection slip submitted.

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

3-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What did you set out to make/learn, and what did you actually end up with? 2. What’s the most surprising thing you discovered while making this? 3. Where will you use Nikhilam in your life outside this classroom?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can write a 1-paragraph “what would Module 11 look like for Grade 4?” — input for Primary band.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark quiz immediately after class while presentations are fresh.
- If presentations spill into Day 11, protect that time.
- The “reflection on Nikhilam in life” question (oral) — listen for genuine answers (“I’ll use it on price tags in the shop”; “I’ll use it for mental estimation”). The honest answer “I won’t use it much” is also valid.

Citation(s) used in this lesson

- All sources verified across days 1–9.